Project: reservoir computing for connectome networks I

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The rate model tries to compute the firing rate (in spikes per second) of each single neuron as a continuous variable. Thus, the i-th neuron has a rate $r_i(t)$ at time t.

The evolution of the model with N neurons is dictated by the differential equation

$$\frac{dr_i(t)}{dt} = -r_i(t) + \phi\left(\sum_{j=1}^{N} A_{ij} r_j(t)\right), \tag{1}$$

where the sigmoidal function

$$\phi(x) = \frac{1}{1+\exp(-x)} - \frac{1}{2} \tag{2}$$

converts the input current of the neuron to rate. Using an Euler integrator, I propose you to integrate the differential equation in the computer and study the behaviour of the system for different topologies of the network. This can be done by plotting individual trajectories and analysing the statistics of the mean firing rate, $r(t) = N^{-1} \sum_{i=1}^{N} r_i(t)$. For example, one can compute the mean and variance over time of $r(t)$ after letting the network enough time to reach an stationary state. To integrate the differential equations, one needs an initial condition. It suffices to draw random values for a small interval of rates (e.g. $r_i = [0, 1]$).

1. Study the case in which the network is fully connected and all links have the same weight, $A_{ij} = J/N$ for all i, j.
2. See what happens when nodes are drawn randomly with a probability p. In this case it is useful to control the number of connections, $k = pN$, so $A_{ij} = J/k$ with probability p and 0 otherwise. This is known as the Erdos-Renyi network.
3. Finally, studying the Gaussian random network is instructive. In this fully-connected network, each element is drawn randomly from a Gaussian distribution, as $A_{ij} \sim N(0, J^2/N)$.
4. We will add the data just by setting $A_{ij} = JD_{ij}$, being D_{ij} the network data and J an arbitrary scaling factor.

Networks 1-3 will serve as control reservoirs to compare to the data later on.